Predicting Wildfire Occurrence Across California

Jeffrey Huber, Sofia Kamage, Nathan Reyes
San Diego State University, San Diego, CA, 92115

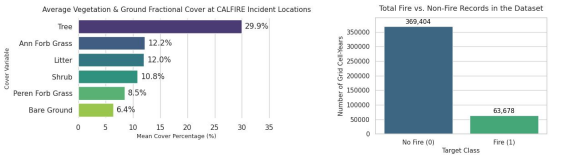
INTRODUCTION

California's wildfire season is growing longer and intensifying under shifting climate conditions. Accurately predicting risk factors for ignition is critical for emergency preparedness and resource allocation. Current risk assessment models rely on static landscape characteristics (e.g. multi-year LANDFIRE data) and overlook dynamic vegetation changes that drive ignition flammability (Brinkmann et al., 2022). This study addresses this limitation by integrating temporal vegetation factors (Rangeland Analysis Platform), high-resolution weather metrics (gridMET), 3D topography factors (LANDFIRE), and traditional at-risk areas as identified by the US Forest Service.
Research Question: To what extent can daily binary wildfire occurrence (Y = 1 for verified ignition ≥ 0.1 acres, Y = 0 for absent non-fire coordinates) be predicted across California?

DATA SOURCES

Data Collection & Feature Engineering

- **Wildfire Incidents (USFS)**: Wildfire points of origin from *National USFS Fire Occurrence Point (Feature Layer)*. The extracted sample consists of N = 535,700 wildfire coordinate observations across California. Key extracted variables include latitude, longitude, final burn acreage, and discovery date and time (24-hour PST).
- **Daily Weather (gridMET)**: Daily surface weather and fire danger matched to coordinate points and date of the wildfires. Key extracted variables include daily maximum and minimum temperatures and relative humidities (tmax, tmin, rmax, rmin), wind speed (vs), solar radiation (srad), daily mean energy release (erc), and dead fuel moisture (fm100, fm1000).
- **Vegetation (RAP)**: Continuous dynamic fuel characteristics were obtained from Rangeland Analysis Platform's (RAP) 30m vegetation cover product. For each observation point, six ground-cover variables were extracted: tree, shrub, annual forb/grass, perennial forb/grass, litter, and bare ground percentages.
- **Zoning & Season (CAL FIRE)**: To model regional climates, each observation was mapped to one of 85 official CAL FIRE seed zones based on similar elevation and climate characteristics. Additionally, a binary seasonal feature (is_peak_season) was created based on CAL FIRE's defined California peak wildfire season (Peak = May-October, Adjacent = November and April, Off-Season = December-March).



The bar plot (left) shows the average vegetation type found at the wildfire incidents. The most frequent vegetation type found at the incidents in the dataset was tree, followed by grass.
The count plot (right) shows the distribution of fires versus non-fires in the dataset.

METHODOLOGY

- **Target Variable (Y):** Binary variable created on ~5 km x 5 km grids annually across California through 2000-2024. Y = 1 if at least one verified USFS fire greater than 0.1 acres occurred within cell grid during that year. Y = 0 for non-fire cells.
- **Train/Test Split:** Time-based partition was used to split observations into training (2000-2018), validation (2019-2021), and test sets (2022-2024).

MODELING & RESULTS

Modeling Pipeline

- **Naive Baseline:** Predicts no fire (Y = 0) for all grid cells. Establishes the benchmark under extreme class imbalance in fire occurrences.
- **Baseline Logistic Regression:** Standard logistic regression pipeline using default probability threshold (P ≥ 0.5) and unweighted loss.
- **Tuned Logistic Regression:** Logistic regression using class balancing (class_weight = 'balanced') paired with optimal F1 decision threshold.
- **Baseline Multi-Layer Neural Network:** PyTorch feedforward MLP (input → 128 units → 64 units → 1 output) with ReLU activations, trained without class scaling, 30% dropout and an F1-optimal decision threshold, used to test whether weighting the rare fire class improves performance.
- **Class-Weighted Multi-Layer Neural Network:** Same architecture with a class-weighted loss (pos_weight = r, 2r, or √r)

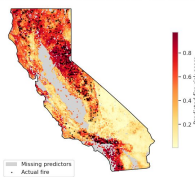
Model	pos_weight	Threshold	Precision	Recall	F1	ROC-AUC	PR-AUC	PR-AUC / naive	TP	FP	FN	TN
Naive Baseline	N/A	N/A	0.000	0.000	0.000	0.500	0.035	1.00x	0	0	1,870	52,058
Baseline Logistic Regression	1 (none)	0.50	0.094	0.031	0.047	0.750	0.080	2.31x	58	557	1,812	51,501
Tuned Logistic Regression	4.651	0.60	0.081	0.495	0.139	0.750	0.081	2.33x	925	10,518	945	41,450

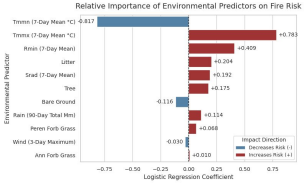
Model	pos_weight	Threshold	Precision	Recall	F1	ROC-AUC	PR-AUC	PR-AUC / naive	TP	FP	FN	TN
MLP (baseline)	1 (none)	0.22	0.121	0.585	0.200	0.794	0.134	2.91x	1,081	7,872	768	30,725
MLP v2.1	$r = \frac{\text{non-fire cells}}{\text{fires}} = 4.7$	0.55	0.113	0.619	0.191	0.789	0.128	2.78x	1,144	8,982	705	29,615
MLP v2.2	2r	0.74	0.116	0.593	0.193	0.784	0.116	2.52x	1,097	8,394	752	30,203
MLP v2.3	√r	0.38	0.117	0.611	0.196	0.791	0.130	2.83x	1,129	8,521	720	30,076

Effect of Class Weighting on Multi-Layer Network (MLP)

We compared four versions of the MLP that differ only in the class weight applied to fire cells in the loss (pos_weight): none (baseline), r, 2r, and √r, where r = (non-fire cells) / (fire cells) = 4.7 in the training years. Each model's decision threshold was chosen to maximize F1 on the validation years (2019–2021), and all were evaluated on 2022–2024. All four models performed similarly: PR-AUC of 0.116–0.134 (2.5–2.9× the naive rate of 0.046), ROC-AUC of about 0.79, and about 60% of burned cells detected. The unweighted baseline scored highest overall (PR-AUC 0.134, F1 0.200), and doubling the weight (2r) performed worst. A larger weight mainly pushed the predicted probabilities up, so the selected threshold rose with it (0.22 → 0.74) and largely cancelled the weighting's effect. Tuning the threshold on held-out data handled the class imbalance as well as reweighting the loss

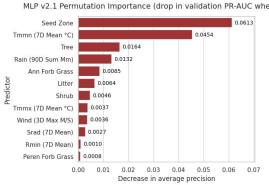
The choropleth map (right) shows the spatial distribution of MLP v2.1 predicted fire occurrence probabilities across California for the test year 2024. Actual verified USFS wildfire locations (black dots) show strong alignment with the predicted high-risk zones (P ≥ 0.7). Gray regions represent urban and agricultural cells masked out by RAP vegetation data.





The horizontal diverging bar chart (left) shows the standardized coefficients of the tuned logistic regression indicating feature impact on the log-odds of fire occurrence. Positive values (red) indicate increased predicted risk, while negative values (blue) indicate decreased predicted risk. Temperature factors (Tmmx; +0.783 and Tmmin; -0.817) demonstrated the strongest magnitude of impact on predicted fire risk.

The horizontal bar chart (right) shows the relative feature importance of MLP v2.1 factors on wildfire risk, measured by decrease in validation average precision (PR-AUC). Spatial zoning (Seed Zone; 0.0613 drop) and mean 7-day temperature lows (Tmmn; 0.0454 drop) show the strongest overall impact on predicted wildfire risk.



CONCLUSIONS

Takeaways

- Modeling California lands on a seasonal basis aggregated in ~5 km grids provides actionable insight into risk factors that can help with resource allocation.
- Together, similar vegetation features (categorized by seed zones) and drought stress factors (gridMET; minimum and maximum temperatures) provide the greatest predictive power to annual ignition risk.

Limitations

- Model input relies strictly on biophysical weather and vegetation data, future studies should include human driven ignition factors such as housing density, campgrounds, powerlines, and roads.
- Aggregating daily gridMET weather into seasonal averages masks severe daily weather shocks such as Santa Ana or Diablo winds that drive ignitions outside of peak wildfire season, highlighting the need for regional models that incorporate specific regional features.

REFERENCES

1. Brinkmann, P., et al. (2022). *Catastrophe models for wildfire mitigation: Quantifying credits and benefits to homeowners and communities*. Casualty Actuarial Society Research Paper.
2. Abatzoglou, J.T. (2013) Development of Gridded Surface Meteorological Data for Ecological Applications and Modelling. International Journal of Climatology, 33, 121-131. https://doi.org/10.1002/joc.3413
3. Jones, M. O., B. W. Allred, D. E. Naugle, J. D. Maestas, P. Donnelly, L. J. Metz, J. Karl, R. Smith, B. Bestelmeyer, C. Boyd, J. D. Kerby, and J. D. McIver. 2018. *Innovation in rangeland monitoring: annual, 30 m, plant functional type percent cover maps for U.S. rangelands, 1984–2017*. Ecosphere 9(9):e02430. 10.1002/ecs2.2430
4. Rollins, Matt. (2009). LANDFIRE: a nationally consistent vegetation, wildland fire, and fuel assessment. International Journal of Wildland Fire. 18. 235-249. 10.1071/WF08088.
5. U.S. Forest Service. (2020). *National USFS Fire Occurrence Point (Feature Layer)*. USDA Forest Service Enterprise Geospatial Data Discovery.